



A Data-Driven Exploration of the King County Housing Market

A Client-Centered Analysis of Luxury Waterfront Homes

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The Analytical Journey

01 — Client & Goals

02 — Data Journey

03 — Search Criteria

04 — Hypothesis Results

05 — Timing & Resale

06 — Recommendations

01 Client & Goals



Meet
Jennifer Montgomery

Jennifer is a high-budget buyer looking for a distinctive property in King County.



- Waterfront
- High building grade
- Renovated
- High budget
- Purchase within 1 month
- Potential resale within 1 year



Can the data help Jennifer identify the right property and understand its value, location, timing and resale potential?

01 Client & Goals



From Client Needs
to Data Questions

Needs

 **Waterfront**
 **High grade**

H1. Is there a premium when waterfront + high grade occur together?

 **Renovated**
 **Older luxury home**

H2. Does renovation add value among older high-grade homes?

 **Location**
 **High budget**

H3. Where are high-grade waterfront homes located and how does value vary?

 **Buy in 1 month**

How frequently do suitable homes appear in the historical sales data?

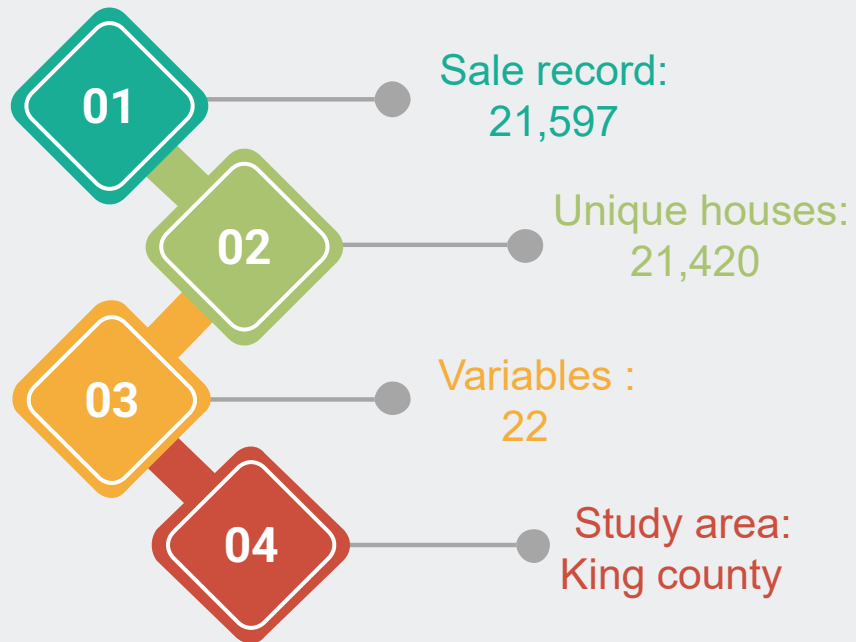
 **Resell in 1 year**

What does short-term resale evidence tell us?

Data
Questions

02 Data Journey

Understanding the Dataset



What does the data describe?



PostgreSQL → Joined Data → Python / pandas



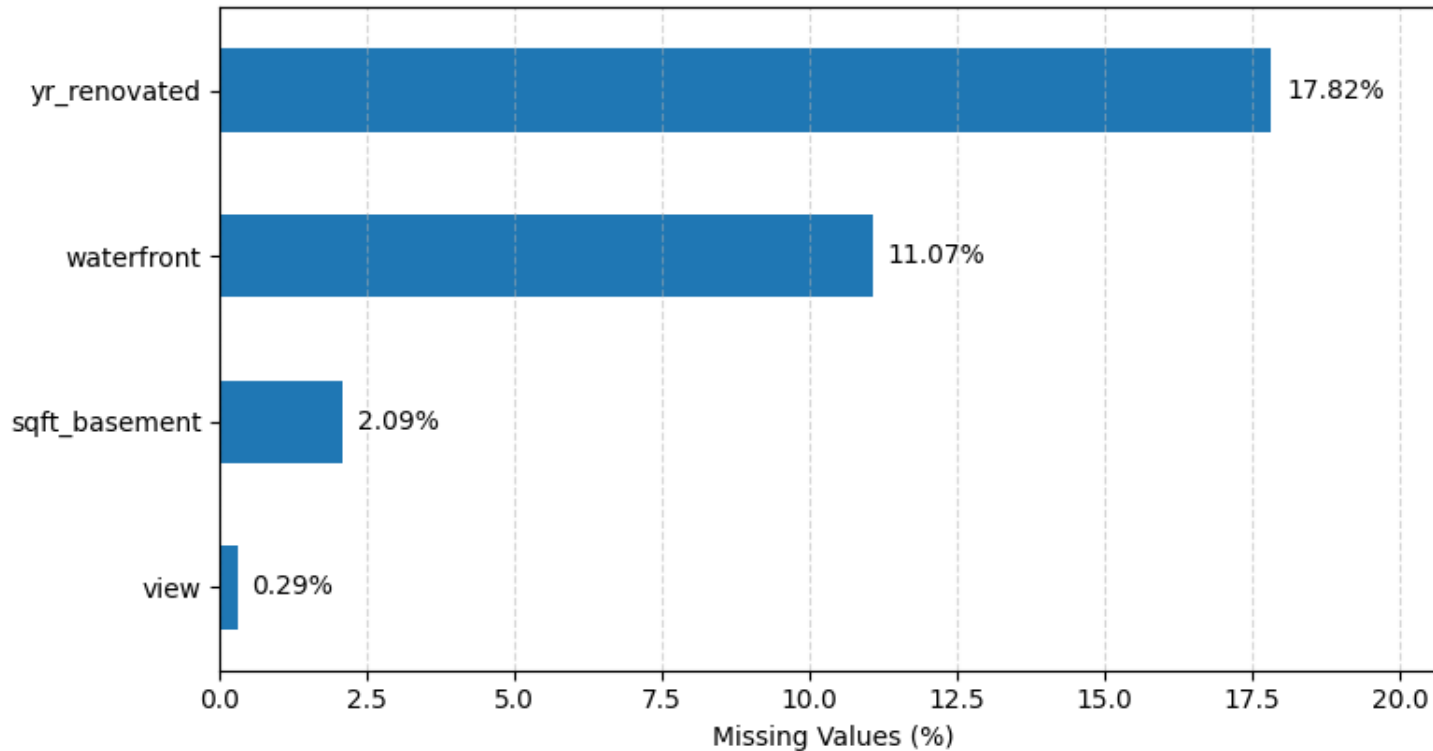
02 Data Journey

Investigating
Data Quality

Completeness



Missing Values in the Dataset



Missing $\neq$ No

Unknown values were preserved
rather than interpreted as zero.

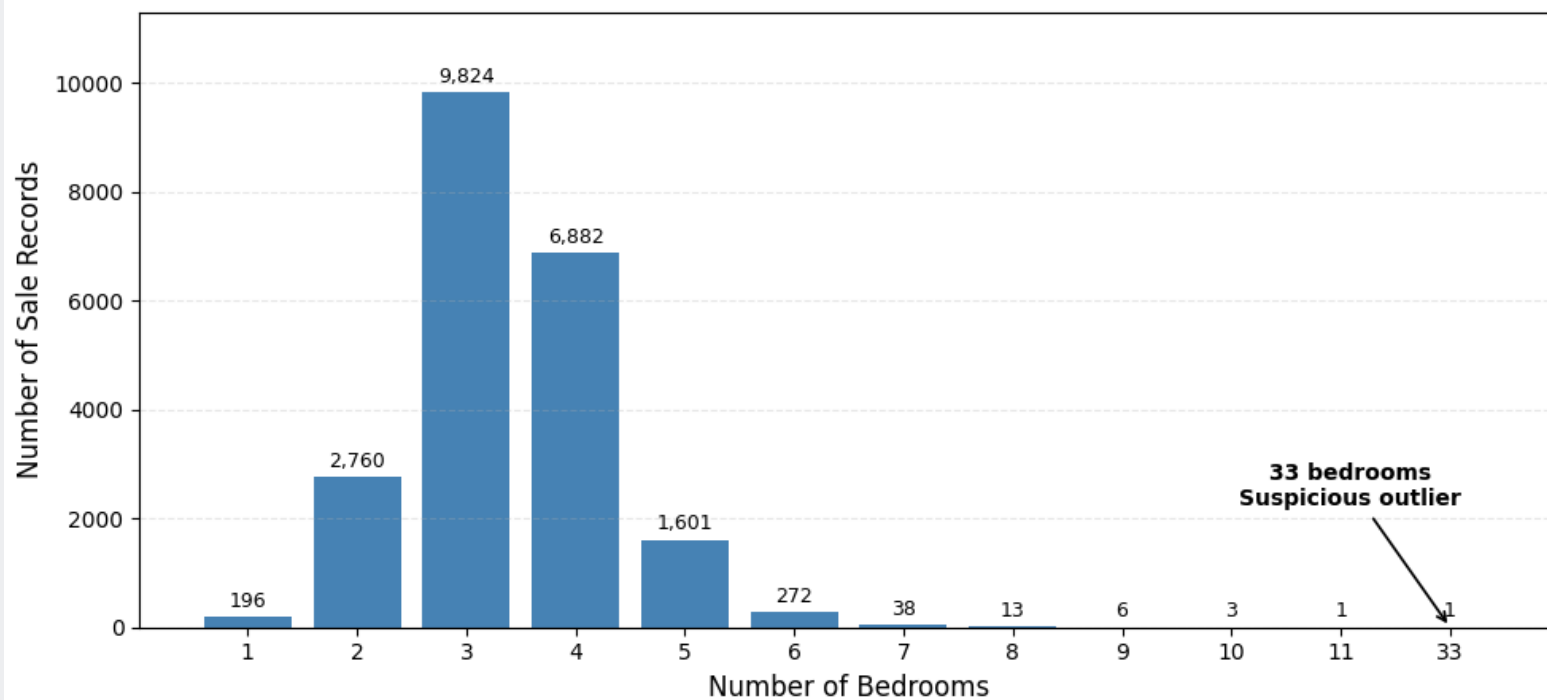
02 Data Journey

Investigating
Data Quality

Plausibility



Bedroom Distribution — Before Cleaning



Most houses: 2–5 bedrooms



**33 bedrooms
Suspicious value**

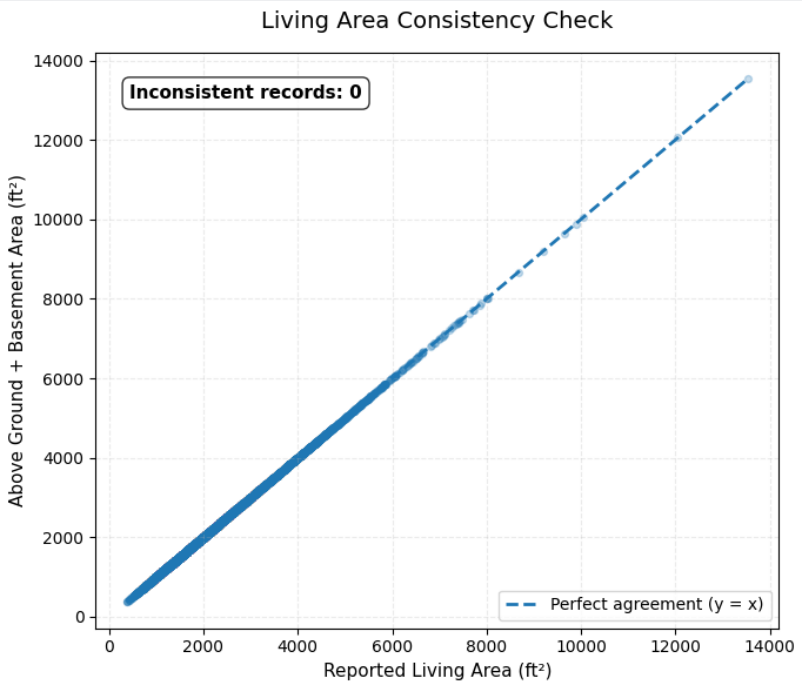
02 Data Journey

Investigating
Data Quality

Consistency, Years & Spatial Validation



Internal consistency



Renovation Year Check

Check	Result
Positive records	744
Raw range	19,340–20,150
Ending in 0	744 / 744
Valid range after correction	1934–2015

Spatial Check

Check	Result
Latitude	47.16–47.78
Longitude	–122.52– –121.32
CRS	EPSG:4326
Spatial extent	Valid for King County

02 Data Journey

Cleaning & Validation Decisions



Issue found	Decision	Final treatment
Renovation year (e.g.19910)	Systematic trailing zero confirmed	19910 → 1991
Missing waterfront	Missing ≠ “No”	Keep as NaN
Missing renovation year	Missing ≠ “Not renovated”	Keep as NaN
Sale date stored as text	Convert to proper date type	datetime
33-bedroom record	Implausible; true value unknown	Set bedrooms to missing
Coordinate display problem	Source coordinates were valid	Preserve original coordinates

✓ 0 renovations before construction
✓ 0 renovations after 2015

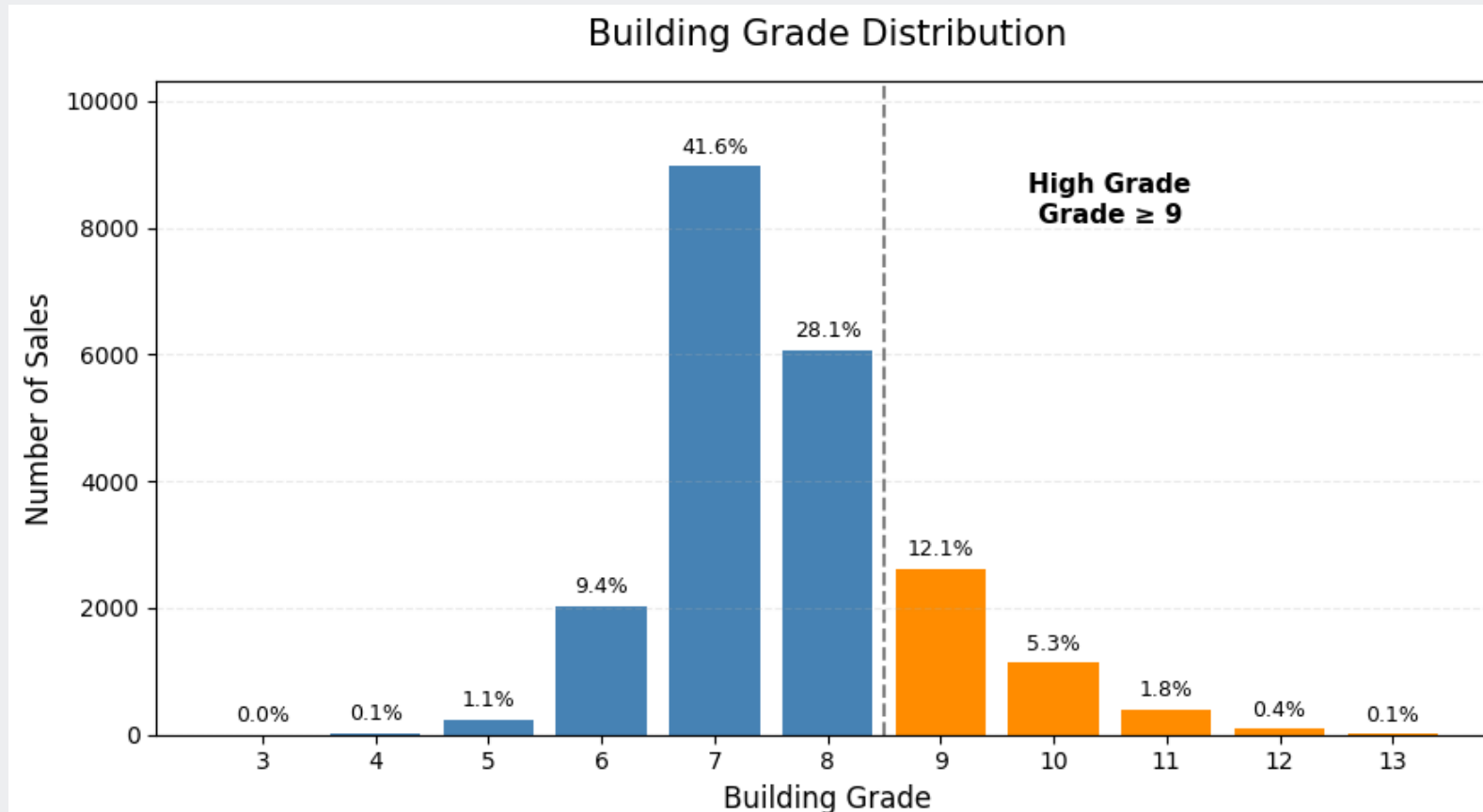
33 BEDROOMS
1,620 ft²
≈49 ft² / bedroom

MISSING ≠ NO
0 = known absence
NaN = unknown

→ set to missing

03 Search Criteria

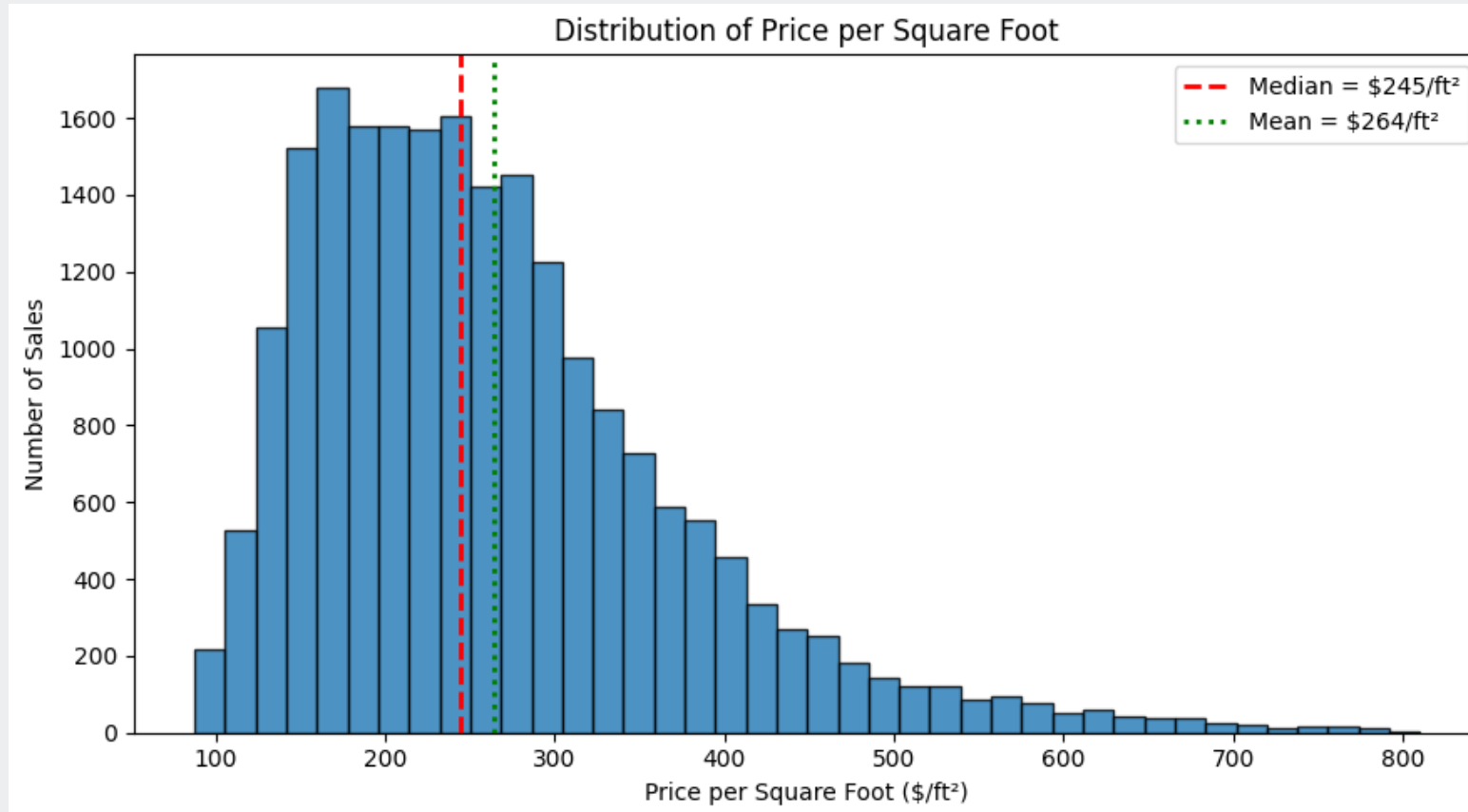
Translating Jennifer's Preferences into Data



High grade: Grade ≥ 9
 $\approx 19.7\%$ of sales

03 Search Criteria

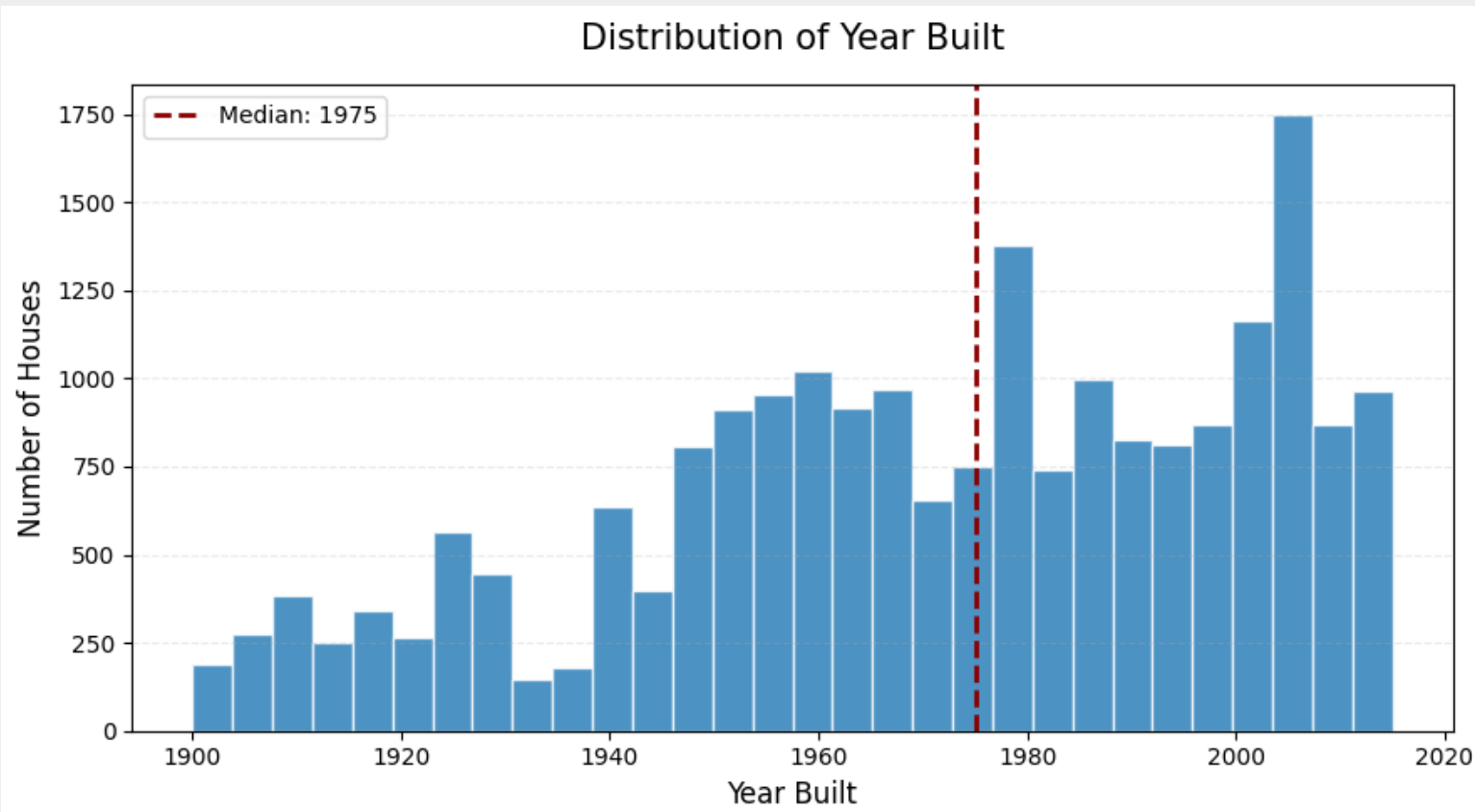
Translating Jennifer's Preferences into Data



This allows properties of different sizes to be compared more fairly.

03 Search Criteria

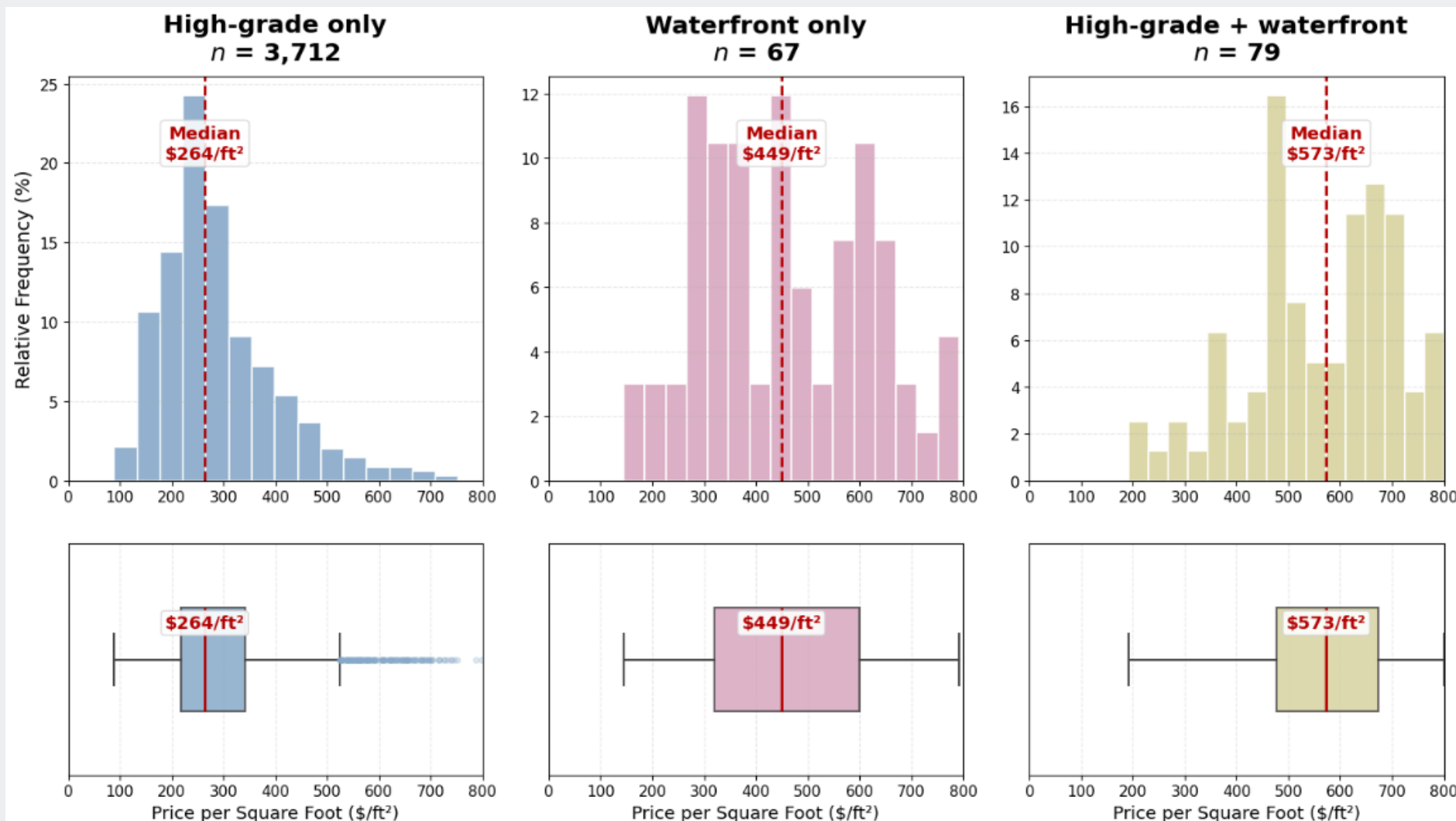
Translating Jennifer's Preferences into Data



Median = 1975 (built year)
Built Year $\leq$ 1975
as older house

04 Hypothesis Results

H1. Does Waterfront + High Grade Create a Premium?



Outcome:

\$573/ft²

Highest median value

+117%
vs high-grade only

+28%
vs waterfront only

Strongest observed premium

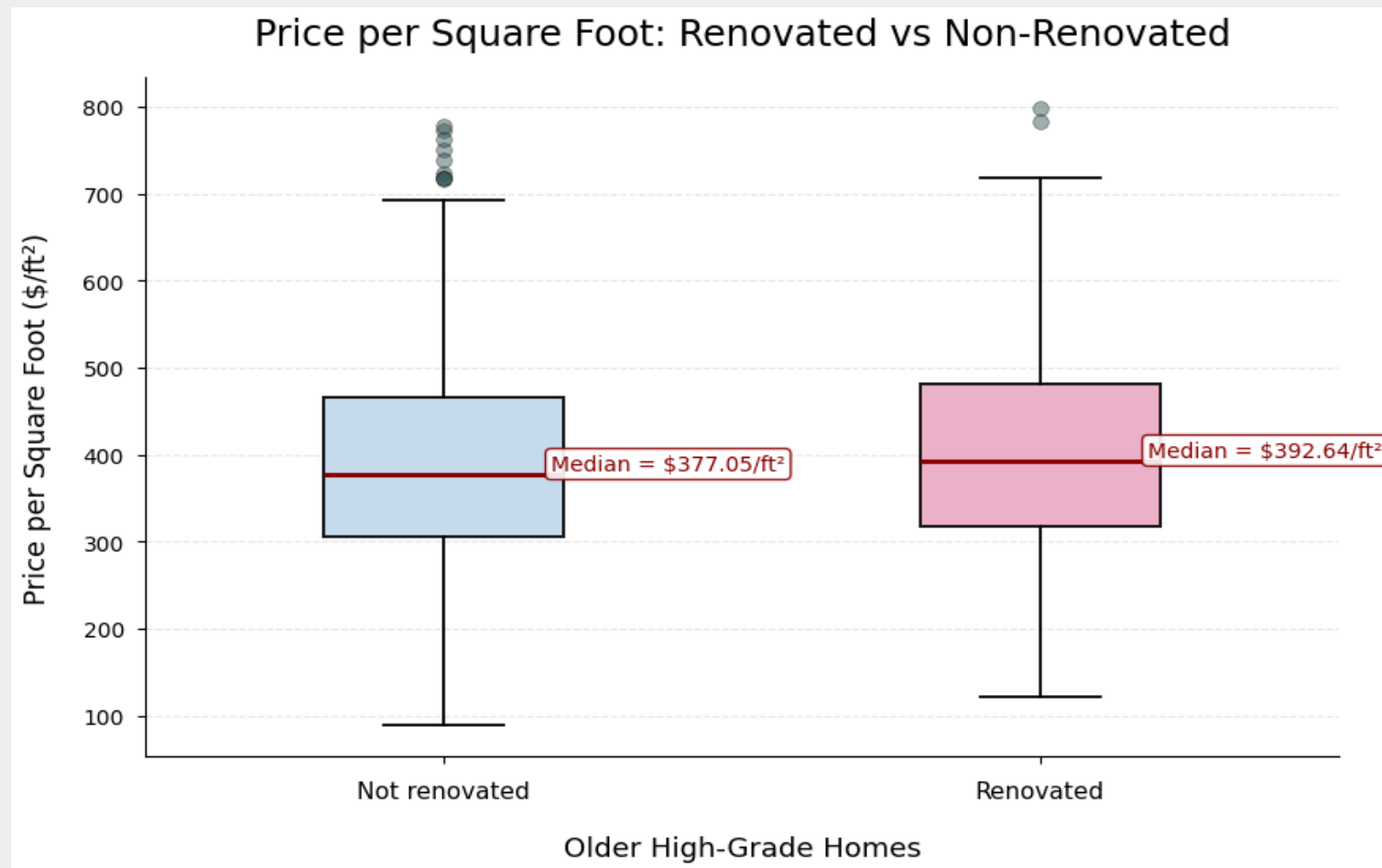


04 Hypothesis Results

H2. Does renovation add value among older high-grade homes?



Older = built ≤ 1975 | High grade = grade ≥ 9



Outcome:

\$392.64/ft²
Renovated median

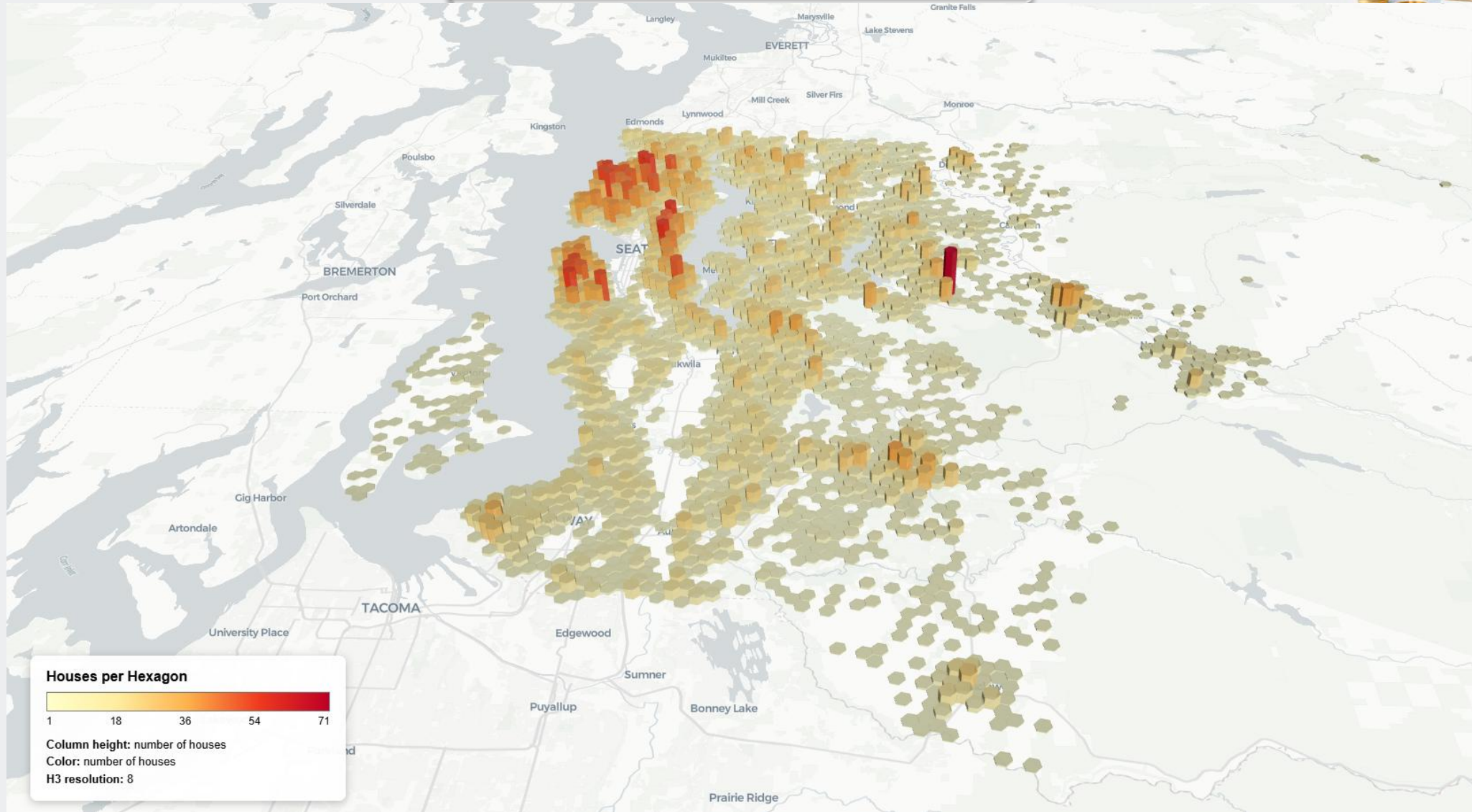
+4.1%
vs non-renovated

Modest observed premium



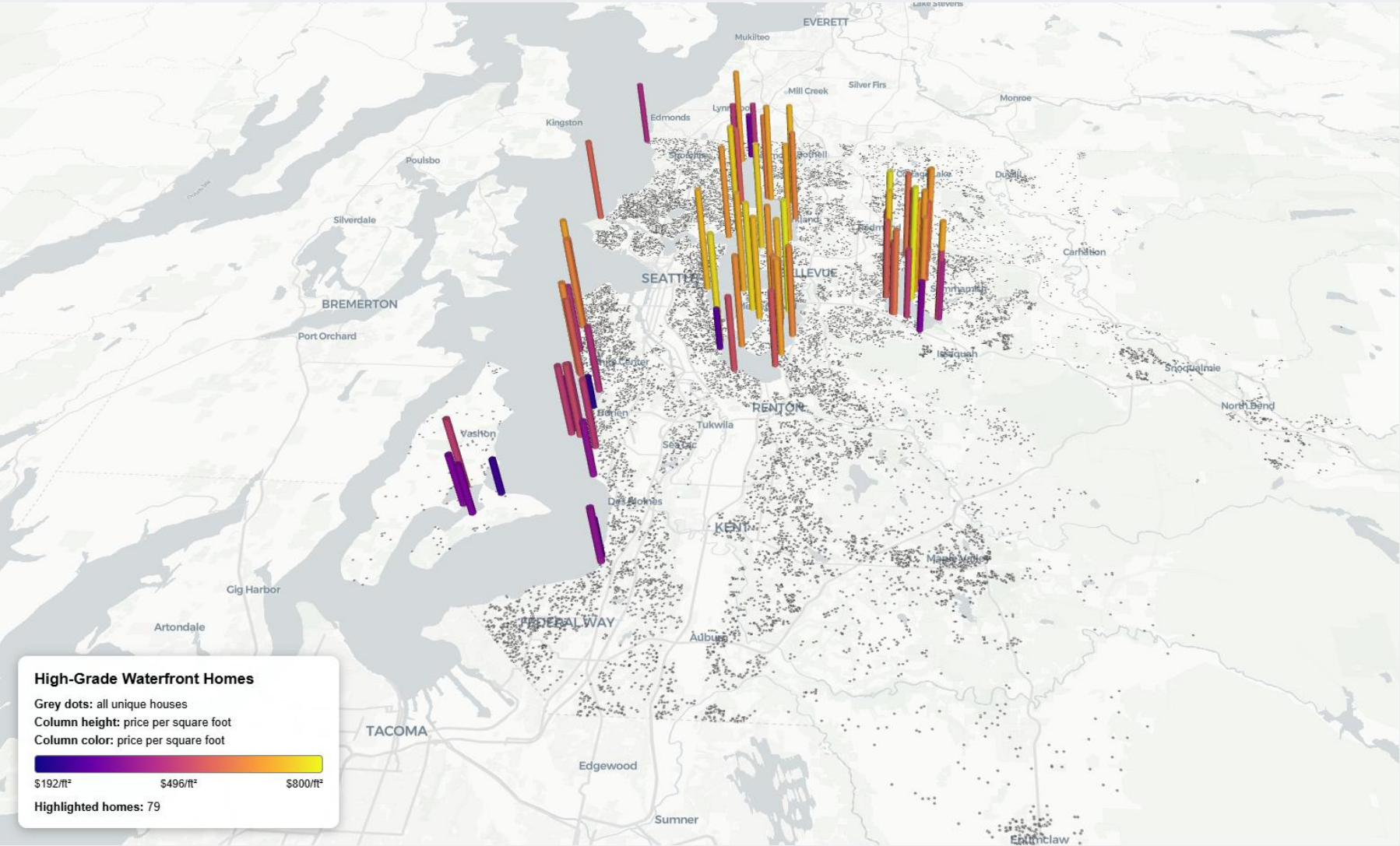
04 Hypothesis Results

H3. Understanding the Spatial Market



04 Hypothesis Results

H3. Where are high-grade waterfront homes located and how does value vary?



Target:
Grade ≥ 9 + Waterfront = 1

Outcome:

79 homes
High-grade + waterfront

Top 4 ZIP codes
≈ 40.5% of target homes

Top 6 ZIP codes
≈ 50.6%

98040 — 9 homes
98166 — 9 homes
98075 — 7 homes
98008 — 7 homes

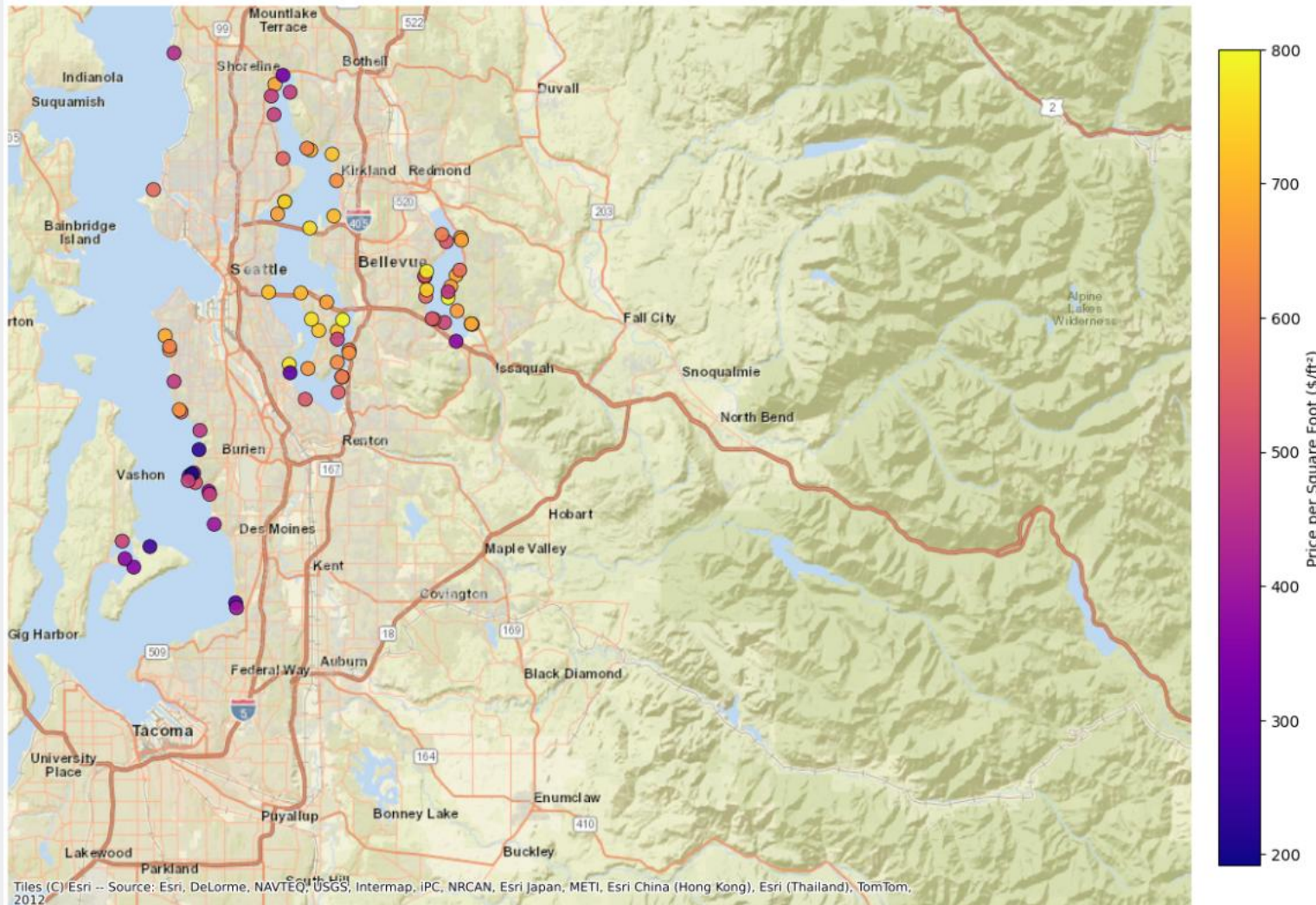
Part-1

04 Hypothesis Results

H3. Does Location Change Price per Square Foot?



High-Grade Waterfront Homes in King County



Outcome:

Strong location effect

\$705/ft²

98040 median

\$394/ft²

98166 median

\$361/ft²

98070 median

Value varies strongly by location

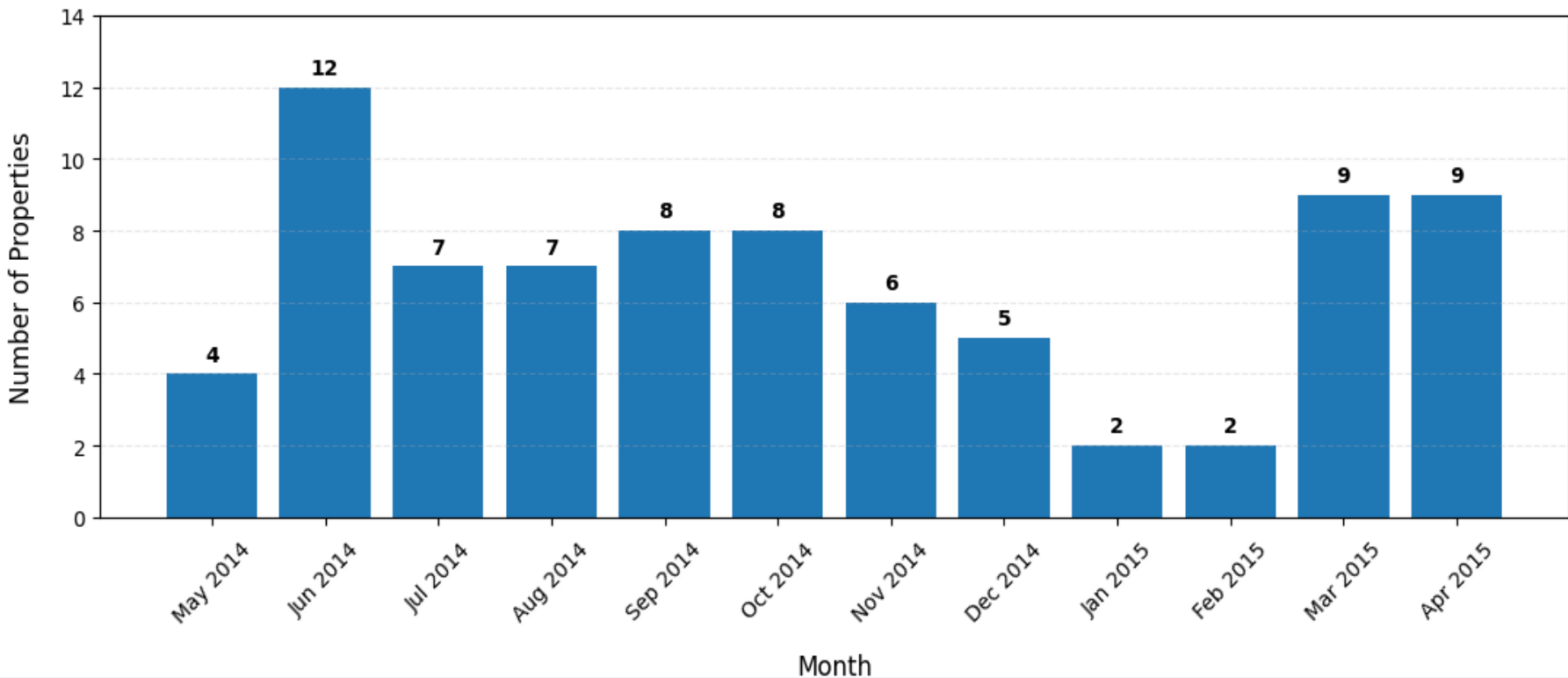
Part-2

05 Timing & Resale

Can Jennifer Find a Suitable Home Within One Month?



Monthly Availability of High-Grade Waterfront Homes



- Outcome:**
- 6.6 homes/month: average activity
 - 2–12 homes: monthly range
 - Every month: suitable sales observed
 - Historically feasible

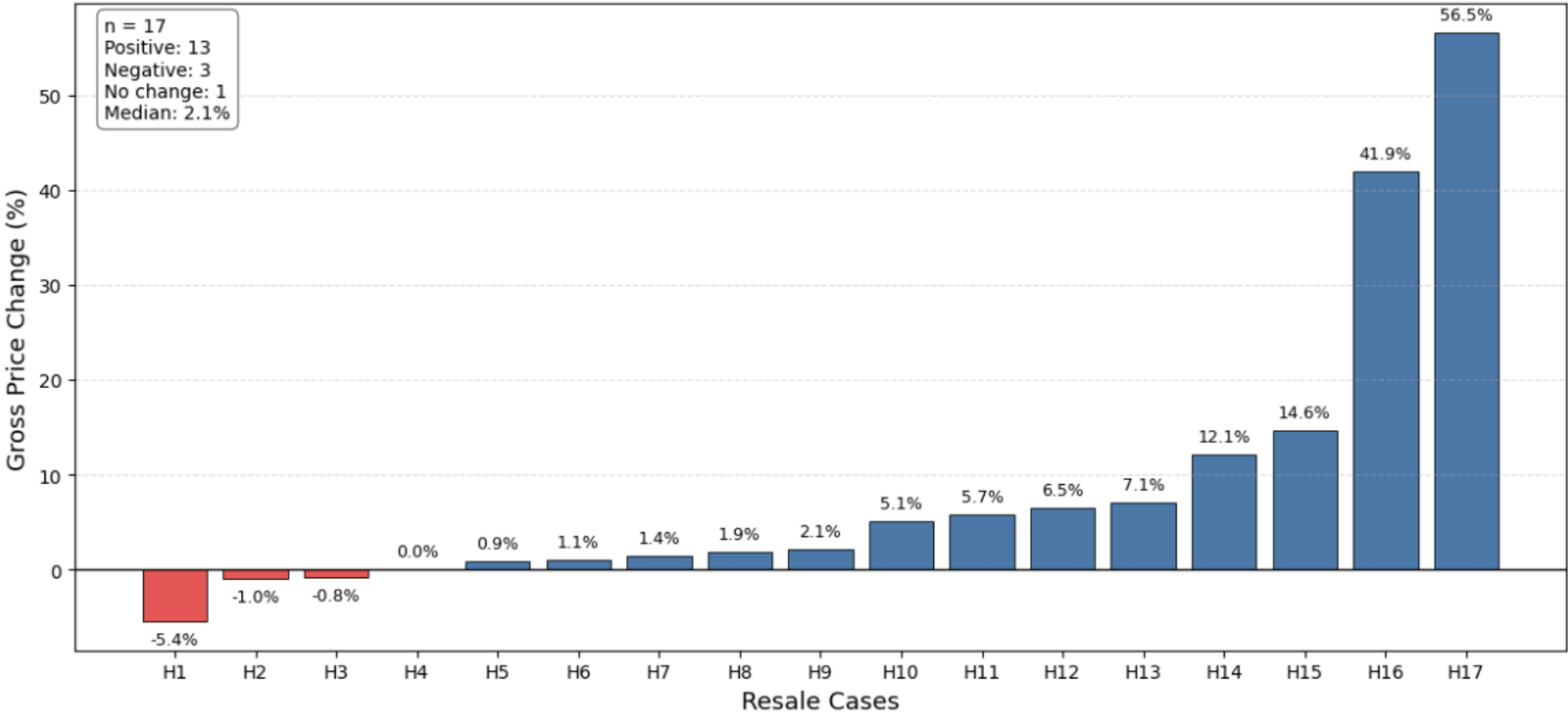


05 Timing & Resale

What Does the Data Say About Resale Within One Year?



Gross Price Change for High-Grade Homes Resold Within One Year



Outcome:

- 17 high-grade resales
- 13 / 17 increased in price
- +2.1% median gross change
- 212 days median holding period
- ⚠ NO DIRECT MATCH
- 0 waterfront short-term resales

06 Recommendations



What Did We Learn for Jennifer?



Property	Renovation
\$573/ft ² — highest median value	+4.1% — modest observed premium
High grade + waterfront first	Renovation useful, but secondary



Recommended Search Strategy:

1. High grade + waterfront first
2. Focus on target locations
3. Renovation second
4. Stay flexible on timing
5. Do not assume a one-year return



Location	Timing & Resale
79 target homes	2–12 homes/month
40.5% in top 4 ZIP codes	Historical activity observed every month
Focus the search geographically	0 waterfront short-term resales

06 Recommendations



From 21,597 Sales to One Client Decision

Location • Timing • Resale



Jennifer's Best-fit Profile:

- High grade + waterfront
- Renovation as secondary
- Focused geographic search
- One-month window plausible
- Resale evidence limited



Thank You